

## 5. ARES: The Mars Port (Phase 4)

*Mars atmosphere is 0.6% of Earth's. The hard parts of the Earth machine are nearly free there.*

The tube holds back 0.6% of the pressure the Earth tube fights; the plasma window's job is 99.4% done by the planet; exit heating is a fraction of the Earth case. Dust storms, not aerodynamics, are the main environmental design driver.

| Element      | Figure                      | Notes                                                                                 |
|--------------|-----------------------------|---------------------------------------------------------------------------------------|
| Orbit track  | ~3.4 km/s → ~180 km @ 3.3 g | 105-second run to low Mars orbit                                                      |
| Escape track | ~5.0 km/s → ~390 km @ 3.3 g | 155-second run; between Moon and Earth in scale, far easier than Earth in engineering |
| Energy       | 1.6 / 3.5 kWh per kg        | orbit / escape                                                                        |
| Staging      | Phobos & Deimos             | Catch points and propellant depots — the warehouse district; SEP tugs fly the cruise  |

Every Mars mission arrives needing return propellant. ISRU solves it on the surface (Sabatier methalox from CO<sub>2</sub> and water ice); ARES solves it in orbit — propellant rides to Phobos at electromagnetic cost and waits in a depot for any departing vehicle. Return cargo, manufactured exports, and samples ride the same track.

**ARES makes Mars a place you can leave as easily as you arrive.**

**The Moon as deep-space launch platform.** A Mars-transfer from the lunar surface costs ~3.0 km/s (≈1.25 kWh/kg) — roughly one-fifteenth the energy of the same trip from Earth's surface. Anything Mars-bound that can be built on, or staged through, the Moon ships from the lunar track: relay satellites, sensor networks, pre-positioned supplies. Starship carries crew and cargo that must arrive fresh; the mass drivers send everything else ahead, early and cheap. The Mars supply chain runs through the Moon, not around it. ARES is Phase 4: it inherits a mature vacuum-world kit (SELENE, Phase 1; fleet, Phase 2–3) and proven atmosphere subsystems (Chimborazo, Phase 2–3).

## 6. The Build Order

*The lightning flashes before the thunder is heard: SELENE launches silently in vacuum before BRONTE shakes the air on Earth.*

| Phase | System                                                      | What gets built                                                                                                                                                    | What it proves / earns                                                                                                                                                                                             |
|-------|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | <b>SELENE PoC</b><br>(Moon)                                 | 1–3 km surface track, 50–100 g, south-pole ridge                                                                                                                   | 2.9 km @ 100 g clears lunar escape; exports water/regolith/metal to cislunar buyers from day one; grows to 10–12 km (30 g) then ~50 km (3 g)                                                                       |
| 2–3   | <b>Chimborazo</b><br>(Earth) +<br><b>Vacuum-world fleet</b> | 14 km first build of through-mountain line (tube, plasma window, aero-shell) running in parallel with SELENE-class kits stamped on Phobos, Deimos, Ceres, the Belt | Chimborazo earns DoD test revenue and proves cost-per-km; the fleet deploys proven vacuum-world technology across airless bodies — both run concurrently, the core bet and the Earth demonstrator advance together |
| 4     | <b>ARES</b> (Mars)                                          | ~180 km track, featherweight tube                                                                                                                                  | Mars becomes a port; the freight network closes                                                                                                                                                                    |
| 5     | <b>BRONTE full</b><br>(Earth)                               | Extend Chimborazo west to ~38 km (Mach 12.6); beyond that, the 1,000 km-class 3 g injector if economics prove out                                                  | The endgame — built only on validated cost data                                                                                                                                                                    |

SELENE is ~300× shorter than any full Earth injector and skips every atmosphere subsystem. Chimborazo Phase 1 is ~70× shorter and proves the hard atmosphere parts before any large commitment. They de-risk in parallel.

## 7. How the Network Moves Mass

*Three launchers, one supply chain. Once they exist, moving mass around the inner system mostly costs what running the launchers costs.*

- **Earth sends what only Earth can build** — machines, electronics, medicine. The eventual full injector runs 80 t/day (29,000 t/yr); the Chimborazo demonstrator's job is to earn that machine the right to exist. (Rev 1 wrongly pinned the 80 t/day cadence on the demonstrator itself.)
- **The Moon sends raw mass** — propellant feedstock, shielding regolith, structural metal, fired at 2.4 km/s toward cislunar catch points. At full cadence the cislunar propellant supply stops being constrained by what Earth can lift.
- **Mars closes the return trip** — ISRU methalox, samples, and exports ride ARES to Phobos/Deimos depots on electricity.
- **SEP tugs connect the nodes** — weeks per leg, nearly propellant-free; chemical engines do only final descents. The launchers do all the heavy lifting off every surface.
- **Each node funds the next.** Rockets deliver SELENE's first coils (the last rocket buy the lunar build needs); SELENE exports fund Chimborazo; Chimborazo revenue and data fund the fleet; the fleet builds ARES. Every airless body added makes the whole network richer.

## 8. The Skyhook: How Cost Falls Toward \$5/kg

*A rotating tether catches pods and flings them higher. No propellant. More payload per launch.*

A long tether rotates in low orbit; its bottom tip sweeps backward, moving slower than orbital velocity. A pod that arrives at the tip gets grabbed; half a rotation later it is released up high, moving faster — to the Moon, GTO, or wherever the mission calls. With a 1.5 km/s tip speed, the launcher needs 6.3 km/s instead of 7.8: ~35% less energy, a slower exit (less heating), and the kick stage shrinks from ~20% of pod mass to 2–5% for trim — the freed 15–18% becomes payload. That shift is where \$5/kg comes from.

- **Keeping it up:** electrodynamic reboost (current through the tether against Earth's field — demonstrated physics, no propellant) plus momentum returned by catching descending capsules.
- **The lunar extension:** a cislunar rotovator catches SELENE pellets and redirects them to LLO, L1, or Earth-transfer; an Earth-orbit tether receives. Cargo crosses cislunar space without burning its own propellant at all.
- **Provenance:** momentum-exchange tethers date to the 1970s; the Boeing/Tethers Unlimited HASTOL study (NASA NIAC, 1999–2001) detailed a 600 km tether with 3.5 km/s tip speed catching Mach-10 aircraft-launched payloads. Materials and manufacturing at scale, not new physics.

## 9. If the Full Earth Injector Gets Built

*Not the plan — the vacuum fleet is the plan. But if the fleet works, this is what becomes possible.*

A full Earth injector — ultimately the 1,000 km-class, 3 g, 8 km/s human-rated machine — is the most consequential infrastructure ever proposed. The key point: if SELENE works in vacuum and Chimborazo validates the atmosphere subsystems and the cost-per-kilometer, the full injector stops being a physics question and becomes a construction question.

### What \$5–20/kg to orbit changes

- **Satellites:** launch stops dominating cost — a 2,000 kg satellite rides for ~\$20K at \$10/kg vs the \$83–97M a GPS III mission actually paid. Constellations replace on maintenance schedules.
- **Stations:** ISS-class logistics (~\$3.1B/yr, ~\$1.8B of it transport) collapses ~300×; thousand-person stations become economically normal.
- **Orbital manufacturing:** ZBLAN fiber, crystals, alloys — Flawless Photonics drew ~12 km of ZBLAN on ISS in 2024 (1,141 m in one day vs a 25 m prior record); the candid caveat is that published proof of reproducibly superior fiber is still pending. The queue is real either way, and it is waiting on freight prices.
- **Deep space:** with cheap Earth-to-orbit plus lunar propellant, Mars missions stop being nation-sized budget exercises; the Belt becomes commercially reachable.

**The strategic picture.** The operator of an 80 t/day injector controls the access ramp to space — 29,000 t/yr can replenish a damaged constellation in hours. The host nation stops paying for space access and starts charging for it. The vacuum fleet pays for itself; the Earth injector is what the fleet eventually makes possible.

# 10. Zero New Physics

*Every subsystem has flown, fired, or been demonstrated. The opportunity is assembly, not invention.*

| Building block               | Status — verified June 12, 2026                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EMALS (US Navy)              | Operational on USS Gerald R. Ford since 2017 incl. combat deployment; 484 MJ flywheel storage, 45 s recharge. CVN-79 (Kennedy) completed builder's trials Feb 2026, delivery ~Mar 2027. 2023 AFOSR-filed report recommends evolving EMALS for lunar launch.                                                                                                                                                                                                              |
| Superconducting maglev       | 603 km/h crewed record (JR Central L0, 2015) still stands; Shanghai Transrapid commercial since 2003; Chūō Shinkansen now expected NET ~2034 after the Shizuoka settlement (early 2026).                                                                                                                                                                                                                                                                                 |
| REBCO superconductors        | $T_c \approx 92$ K. All-REBCO records: 26.86 T (ASIPP, 2024) → <b>28.2 T</b> (CHMFL, 2025); all-superconducting record <b>35.1 T</b> (ASIPP, Sept 2025); 45.5 T hybrid (NHMFL, 2019). CFS demonstrated 20 T SPARC-class coils (2021); SPARC assembly underway (first TF coil placed Jan 2026), first-plasma targeted 2026, $Q > 1$ targeted 2027; ARC (Virginia) early 2030s. $J_c > 1,000$ A/mm <sup>2</sup> at 30 T at 4.2 K. Launch LSM needs only 5–10 T at the gap. |
| MHD plasma window            | Invented by A. Hershcovitch (Brookhaven), patented 1995; holds ~9 atm; <b>demonstrated</b> for non-vacuum e-beam welding at cm-scale apertures. Scaling to the Ø4 m bore (2.8× baseline power) is the program's R&D centerpiece — Phase 1 tests it at full bore.                                                                                                                                                                                                         |
| X-59 shock shaping           | First flight Oct 28, 2025; <b>first supersonic flight June 5, 2026</b> (Mach 1.1); Mach 1.4 acoustic points next; XVS camera-vision cockpit flight-proven.                                                                                                                                                                                                                                                                                                               |
| High-g electronics           | Gun-launched guidance routinely qualified to 15,000–20,000 g (Excalibur-class). SELENE's 100 g cargo mode is conservative by two orders of magnitude.                                                                                                                                                                                                                                                                                                                    |
| EM launch-assist market      | Auriga Space: \$12.2M cumulative incl. AFWERX D2P2; hypersonic-test-as-revenue model proven in market; outdoor track (Thor) planned 2026.                                                                                                                                                                                                                                                                                                                                |
| Orbital manufacturing demand | Flawless Photonics: ~11.9 km ZBLAN drawn on ISS (Feb–Mar 2024; 1,141 m/day vs 25 m prior record); Luxembourg production expansion; published performance characterization still pending — flagged honestly.                                                                                                                                                                                                                                                              |
| Lunar mass-driver physics    | O'Neill, NASA Ames studies 1976–77; Mass Driver One demo 1977. SJSU sizing study (2023): 25.4 kg pellets, 2.4 km/s, 10 s cadence, 8.7 MW polar solar, ~1.0 MA per bucket coil.                                                                                                                                                                                                                                                                                           |
| Lunar intent at scale        | SpaceX lunar pivot + mass-driver statements (Feb–Mar 2026). Founder remarks, not a program of record — but the direction of the largest launch company on Earth.                                                                                                                                                                                                                                                                                                         |
| Fusion pull (He-3 context)   | NIF ignition Dec 2022 (3.15 MJ); best shot now <b>8.6 MJ, target gain &gt;4</b> (Apr 2025), ignition repeated 9+ times. Commercial He-3 demand is already contracted: Interlune–Bluefors offtake (Sept 2025) for up to 10,000 L/yr from 2028, prospecting mission 2027.                                                                                                                                                                                                  |

## 11. The Hard Parts

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*A document about a project this large that doesn't name the real problems is salesmanship, not engineering.*

- **Plasma window scaling.** Centimeter demos → Ø4.0 m bore at 2.8× baseline power. First R&D dollar; if it doesn't scale, the muzzle needs a different answer and we must learn that at demonstrator scale, cheaply.
- **Cost per kilometer is unknown.** The \$20–250/kg projection lives or dies on vacuum-rated superconducting track cost. Nobody has published a credible figure. Chimborazo Phase 1 produces the first real one.
- **The El Arenal piers.** 110–590 m A-frames in a 0.4 g seismic zone — beyond Millau's 343 m record. Partial trenching, rock-buttressed spans, or accepting a lower exit are the studied outs.
- **The reserve.** Chimborazo sits inside a protected fauna reserve under a rights-of-nature constitution. The legal path may be harder than the rock. Stated plainly because pretending otherwise would be fatal later.
- **SELENE's first bill is a rocket bill.** First coils, power, and radiators ride to the Moon at today's prices; the starter track must fit one or two heavy-lift flights.
- **Catching is harder than throwing.** Until the first cislunar catch happens, export revenue is theoretical. Wide-cone tugs first, tethers later.
- **Polar ice uncertainty.** Shackleton-specific reserves are unproven (bounded  $\leq 5\text{--}10$  wt% in surveyed regolith). The early business plan must close on regolith, oxygen, and metal even in the lean-ice case. Chang'e-7 data lands within a year.
- **The treaty question.** The Outer Space Treaty is silent on commercial mass drivers; a kinetic launcher is inherently dual-use. That ambiguity buys scrutiny on one side and AFWERX funding on the other. EMALS and GPS both walked this road.

## 12. What KERAUNOS Unlocks

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### Orbital manufacturing

- **ZBLAN fiber** — up to 100× lower theoretical loss than silica; ~12 km drawn on ISS in 2024; customers queue on freight prices. KERAUNOS is the freight price. (Performance proof still pending publication — the bet is stated, not oversold.)
- **Orbital bioprinting** — organs that collapse under their own weight in 1 g; daily return pods turn it from stunt to logistics business.
- **Debris salvage** — thousands of tonnes of aerospace alloy already in orbit becomes ore when tug-hours are cheap.

### Geopolitical position

- **The Space Panama Canal** — the host nation stops paying for access and starts charging for it.
- **Presence as a supply chain** — 80 t/day secures orbital slots and lunar territory as a logistics fact, not a flag.
- **The deterrence dividend** — constellation replenishment in hours, not months.

## Fusion economy

- **The He-3 supply chain.** Lunar regolith holds an estimated (model-based) ~1M tonnes of He-3. D-He3 fusion is **low-neutron** (D-D side reactions keep it from being strictly aneutronic) — far less activation, direct conversion potential. At today's ~\$20M/kg (~\$20B/tonne) market price and SELENE's ~\$5/kg lunar launch cost, delivery economics are absurdly favorable; Interlune's contracted offtakes (Bluefors, 2028–2037) prove demand precedes the fusion era. SELENE isn't just a launcher — it's the fuel logistics for the energy industry after next.
- **The energy loop.** Trackside storage doubles as grid-scale battery between launch windows; the same plant powers debris-sweeping lasers over the corridor. Nothing idles.

Verification note: all external claims in this revision were re-checked against primary reporting and literature on June 12, 2026; all physics re-derived from first principles the same day. Annex A (BRONTE Chimborazo Design Map) is bound after Section 4. The SELENE Design Map (KER-SEL-TRK-002) is issued as a companion sheet.